Adaptive Multistrategy Particle Swarm Optimization for Hyperspectral Remote Sensing Image Band Selection

Yuting Wan, *Member, IEEE*, Chao Chen, *Student Member, IEEE*, Ailong Ma, *Member, IEEE*, Liangpei Zhang, *Fellow, IEEE*, Xunqiang Gong, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— Hyperspectral remote sensing band selection picks out characteristic feature combinations to weaken the strong correlation caused by spectral continuity. However, it is difficult for traditional methods with fixed strategies to search the entire space and make adjustments for the optimization process. Thus, the solutions obtained can be mostly local optima. In this article, a novel adaptive multistrategy particle swarm optimization (PSO) for hyperspectral image remote sensing band selection (AMSPSO_BS) is introduced to obtain a subset solution suitable for classification. The problem is modeled as an effective fitness function, and the quotient of the linear discriminant value and the mean mutual information (LD/MMI) is used to remove the redundancy between bands. The randomly generated solutions are then encoded to form a population, which rely on various particle update strategies (PUS) with different reference positions for updating. During the particle motion, the effect of each strategy on population evolution is considered comprehensively and reflected in the change of selection probability. And the motion parameters are dynamically adjusted to balance the global and local capabilities. Four hyperspectral remote sensing image datasets were utilized to conduct band selection experiments, to confirm the effectiveness of AMSPSO_BS.**

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Yuting Wan is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing and the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China, and also with the Key Laboratory of Mine Environmental Monitoring and Improving around Poyang Lake, Ministry of Natural Resources and the School of Surveying and Geoinformation Engineering, East China University of Technology, Nanchang 330013, China (e-mail: wanyuting@whu.edu.cn).

Chao Chen, Ailong Ma, Liangpei Zhang, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing and the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: chenchao0922@whu.edu.cn; maailong007@whu.edu.cn; zlp62@whu.edu.cn; zhongyanfei@whu.edu.cn).

Xunqiang Gong is with the Key Laboratory of Mine Environmental Monitoring and Improving around Poyang Lake, Ministry of Natural Resources and the School of Surveying and Geoinformation Engineering, East China University of Technology, Nanchang 330013, China (e-mail: xqgong1988@163.com).

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***Index Terms*— Band selection, hyperspectral image, motion parameter, particle swarm optimization (PSO), particle update strategy (PUS), remote sensing.**

I. Introduction

HYPERSPECTRAL remote sensing is a new type of earth observation technology based on hyperspectral imaging spectrometers. They separate electromagnetic waves through dispersive prisms, and then record the corresponding energy in different frequencies to form hyperspectral remote sensing images [1]. Compared with traditional imaging techniques, hyperspectral remote sensing realizes the union of imagery and spectrum [2]. The spatial and spectral features of ground objects are reflected in a 3-D image cube, contributing to more detailed object identification and material distinction [3]. It is the remarkable advantage that has resulted in hyperspectral remote sensing being widely applied in various fields, such as environment protection [4], disaster assessment [5], and other earth observation tasks. However, the spectral continuity of hyperspectral images brings more redundant information. What is worse, the classification accuracy decreases with the limited samples, known as Hughes phenomenon [6]. Therefore, it is significant to characterize hyperspectral images more concisely to obtain equivalent or better classification performance.

To handle the problem, band selection has been accepted as an effective way to preprocess hyperspectral images. The main idea is to pick out a significant and decisive subset from all the complex and high-dimensional bands [7]. Sun and Du [8] conducted a comprehensive review of existing researches on hyperspectral band selection. The work mainly focuses on how to select and divides band selection into six main categories. However, what to select is also a key issue. That is, an effective evaluation model for band selection is necessary as well. Some researchers have made efforts in this field. Li et al. [9] combined the accuracy and the percentage of eliminated bands as metric for better classification precision and smaller size of subset. Sun et al. [10] utilized correntropy to select bands from sparse subspace. He et al. [11] presented a valid band evaluation index to extract the most representative subset.

The typical band selection methods for hyperspectral remote sensing images only pick the dominant bands orderly and do

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not fully take their combinatorial characteristics into consideration [12]. In fact, band selection is a difficult task with search space expanding exponentially. Furthermore, since there is a strong correlation between adjacent bands, the gradient-based methods are more likely to suffer from local stagnation in the multipeak optimization space [13]. Therefore, a more efficient optimization approach is needed to obtain a better subset.

Evolutionary computation (EC) which originated from biological evolution [14] has been proven to be an effective strategy suitable for optimization problems. EC methods tend to initialize a set of solutions and optimize them simultaneously according to certain evolutionary strategies, which contributes to their global optimization potential. This helps with efficient search in the solution space of band selection. And with flexible expression and relaxed application conditions [15], EC methods are capable of dealing with band selection issues [16].

In fact, EC methods were used in relevant applications in the early stage of research into band selection [17], [18]. EC-based band selection methods model the problem as a proper objective fitness function and use "1" and "0" to characterize whether each band is selected or not, to initialize the population. The population is then evolved iteratively through various evolutionary operators. Finally, the optimal individual in the search process is decoded into a band subset. Based on the process, many EC paradigm algorithms have been adjusted and improved for application to band selection. For example, a multilevel optimal classification band selection model based on a genetic algorithm (GA) and rough set theory [19] was proposed as early as 1998. Nagasubramanian et al. [20] also combined the GA and support vector machine (SVM) for the identification of an effective band subset. An improved differential evolution (DE) algorithm [21] has been used for the optimal band subset. What is more, ant colony optimization [22] was applied through two-stage optimization for feature selection. Zhang et al. [23] adopted a clonal selection algorithm to perform the task of dimensionality reduction.

The above evolutionary methods adopt fixed search strategies to remove the redundancy between bands during the optimization, which is "blind" to the evolutionary process. On the one hand, there is a requirement for adjusting the optimization capabilities in the different evolutionary stages [24]. Most solutions in the early population are less optimized for the random initialization approach. It is necessary to encourage individuals to cross the obstacles caused by the nearby gradient and perform global large-scale optimization. In the later stage of the population evolution, most of the individuals will have been optimized to some extent through fitness growth-based iteration. Therefore, meticulous local exploitation instead of cursory global exploration is required to ensure that the high-quality solutions in the neighborhood are not lost [25]. On the other hand, feedback during optimization can facilitate problem-solving since the merits of a given evolutionary direction can be established from past evolution [26].

In addition, the spatial and spectral content of hyperspectral remote sensing images is complex and coupled with the various environmental interferences during acquisition. It is difficult to achieve generalization with a single method, which is called "no free lunch" (NFL) [27].

Therefore, it is vital to make the optimization process more adaptive to the problems themselves. Unfortunately, the optimization mechanisms of most evolutionary algorithms are fuzzy and of high complexity, hard to make any proper adjustments. Particle swarm optimization (PSO), with a clear optimization mechanism and low model complexity, can meet the demands, which makes self-adaptation possible.

In this article, a novel improved adaptive multistrategy PSO is proposed for hyperspectral remote sensing image band selection (AMSPSO_BS) is introduced to obtain a subset solution suitable for classification. The work has the following contributions:

1) To solve the two problems of how to select and what to select comprehensively, a novel hyperspectral image band selection framework is established. The problem is modeled as an effective fitness function—the quotient between the linear discriminant value and the mean mutual information (LD/MMI)—to retain separability for better classification performance while removing redundancy of the selected subset. Optimization is then conducted through the improved PSO method to obtain the optima.
2) A multistrategy evolutionary model is applied in the particle updating to take advantage of the feedback from the past evolution. To handle the rapid convergence caused by the fixed reference, a particle update strategies (PUS) pool is designed for various optimization directions. The optimization process is guided by not only the current optima, but also the average optima and other particles. In addition, a dynamic adjustment method for the selection probability is introduced to make full use of the "experience" in the optimization process, thus determining the dominant evolutionary direction.
3) The motion parameters decrease linearly for adjustment of the optimization capability with the iterative process. The change of particle position depends on the update of the velocity, which is positively related to the inertia weight w and acceleration factors c_1 and c_2. Therefore, the particles will have greater speed to move globally in the early exploration stage, taking advantage of population-based search. While in the later exploitation stage, they are limited to converging to the nearby dominant space as soon as possible and developing local solutions.

The remainder of this article is arranged as follows. Section II introduces the basic principle of PSO. Section III provides some detailed descriptions of AMSPSO_BS. The experimental setup and the results are presented in Section IV. Section V discusses and analyzes the performance of the method further. Finally, Section VI concludes the article.

II. Related Research Background

PSO is one of the representative algorithms of swarm intelligence that was developed in 1995 [28] by simulating the foraging behavior of flocks of birds. Through observations of

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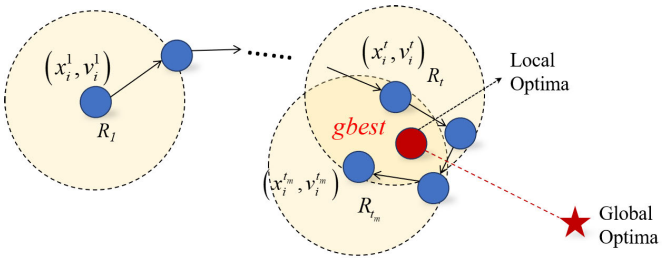


Fig. 1. Main concerns of the traditional PSO method.

the real world, it is assumed that the bird swarm motion model is based on computational manipulation between individuals. All the individuals obtain favorable position information from their neighbors, so that the population moves toward the dominant areas with a high degree of randomness.

Based on the above principles, PSO uses particles with only the properties of position and velocity for optimization problems [29]. The position s is the encoded representation of the solution, while the velocity v refers to the changing trend of the dimensional components of the position. Each particle searches for the optimal solution separately in the search space and records the best position ever reached as the individual optima $pbest_i$. The $pbest_i$ with the highest fitness is the current population optima $gbest$. All the particles adjust their speed and positions according to the current $pbest_i$ they have found and the current $gbest$ shared by the entire particle swarm, as shown in the following equations:

$$v_i^t = wv_i^{t-1} + c_1 r_1 \left(pbest_i - s_i^{t-1}\right) + c_2 r_2 \left(gbest - s_i^{t-1}\right) \quad (1)$$

$$s_i^t = s_i^{t-1} + v_i^t \quad (2)$$

where w is the inertia weight; c_1 and c_2 are the acceleration factors for the $pbest_i$ and the $gbest$, respectively; r_j represents a random fraction in the range [0, 1]; the ordinal i refers to the ith particle in the population; and t is the current number of iterations.

Ideally, $gbest$ would become progressively closer to the global optima, while $pbest_i$ would converge toward the global optima or local optima. However, this is not always the case, and there are two main concerns, as indicated in Fig. 1. In the actual optimization process of hyperspectral image band selection, $pbest_i$ and $gbest$ are likely to be local optima, caused by the huge search space and strong correlation between bands. With the fixed strategies used to improve the particle motion, the particle swarm will converge rapidly around the local optima, easily ignoring search space that is never explored [30]. In addition, the motion parameters are positively related to the velocity and determine the search scope in an iteration. So, particles with fixed motion parameters can fluctuate near the population optima instead of converging to it when the optimization level is high.

Overall, the traditional PSO suffers from unbalanced global and local search capabilities and further decrease solving efficiency [31].

To address the above concerns, AMSPSO_BS is designed with a self-adaptive PUS pool to avoid the quick convergence caused by the fixed reference positions. The motion parameters also change with the iteration, to make sure of the conversion from global exploration to local exploitation. The detailed implementation procedures are presented in Section III.

III. Proposed Method

Due to the imbalance and insufficient optimization abilities caused by the fixed PUS, adaptive multistrategy PSO with dynamic rectification strategies is adopted for the optimal subset selection. An efficient objective function for subset evaluation is first constructed. Subsequently, the randomly initialized solution set is optimized by the novel optimization method. The main workflow is shown in Fig. 2.

A. Optimization Model for AMSPSO_BS

The main purpose of hyperspectral remote sensing image band selection is to maintain the classification performance while removing the redundant information of the subset. Thus, this can be modeled as an optimization problem, with proper parameters used to simultaneously evaluate the category separability and the redundancy.

In this article, an evaluation criterion for the band subset based on information theory was proposed. The evaluation criterion consists of two important parameters: the LD value and MMI. Parameter LD [32] as shown in (3), is the trace of the division result of the interclass scatter matrix S_b (4) and the intraclass scatter matrix S_w (5). It guarantees strong category separability by maximizing the interclass distance and minimizing the intraclass distance to improve the classification performance. Parameter MMI (6) [33] is selected to estimate the amount of the redundant information hided in different bands, which depends on the joint probability (7). It can be calculated through the joint gray level histogram (8). The mean of the mutual information (MI) of any two bands is the MMI. Therefore, the quotient of LD and MMI can assess the category separability and the redundancy of the selected subset effectively. The relevant formulas are given as follows:

$$\mathrm{LD} = \mathrm{tr}\left(S_w^{-1} S_b\right) \quad (3)$$

$$S_b = \sum_{k=1}^{K} P(c_k)(\mu_k - \mu_0)(\mu_k - \mu_0)^T \quad (4)$$

$$S_w = \sum_{k=1}^{K} P(c_k)\mathrm{cov}(c_k) \quad (5)$$

in which $P(c_k)$ is the prior probability and $\mathrm{cov}(c_k)$ is the covariance matrix of the kth class. μ_k refers to the mean spectrum vector, while μ_0 represents the mean of μ_k. And it is assumed that there are K classes in the hyperspectral image.

According to information theory, the MMI of the hyperspectral image is defined as shown in the following equations:

$$\mathrm{MMI} = \frac{1}{C_{n_B}^2} \sum_{i=1}^{n_B-1} \sum_{j=i+1}^{n_B} I\left(B_i; B_j\right) \quad (6)$$

$$I\left(B_i; B_j\right) = \sum_{x \in B_i} \sum_{y \in B_j} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (7)$$

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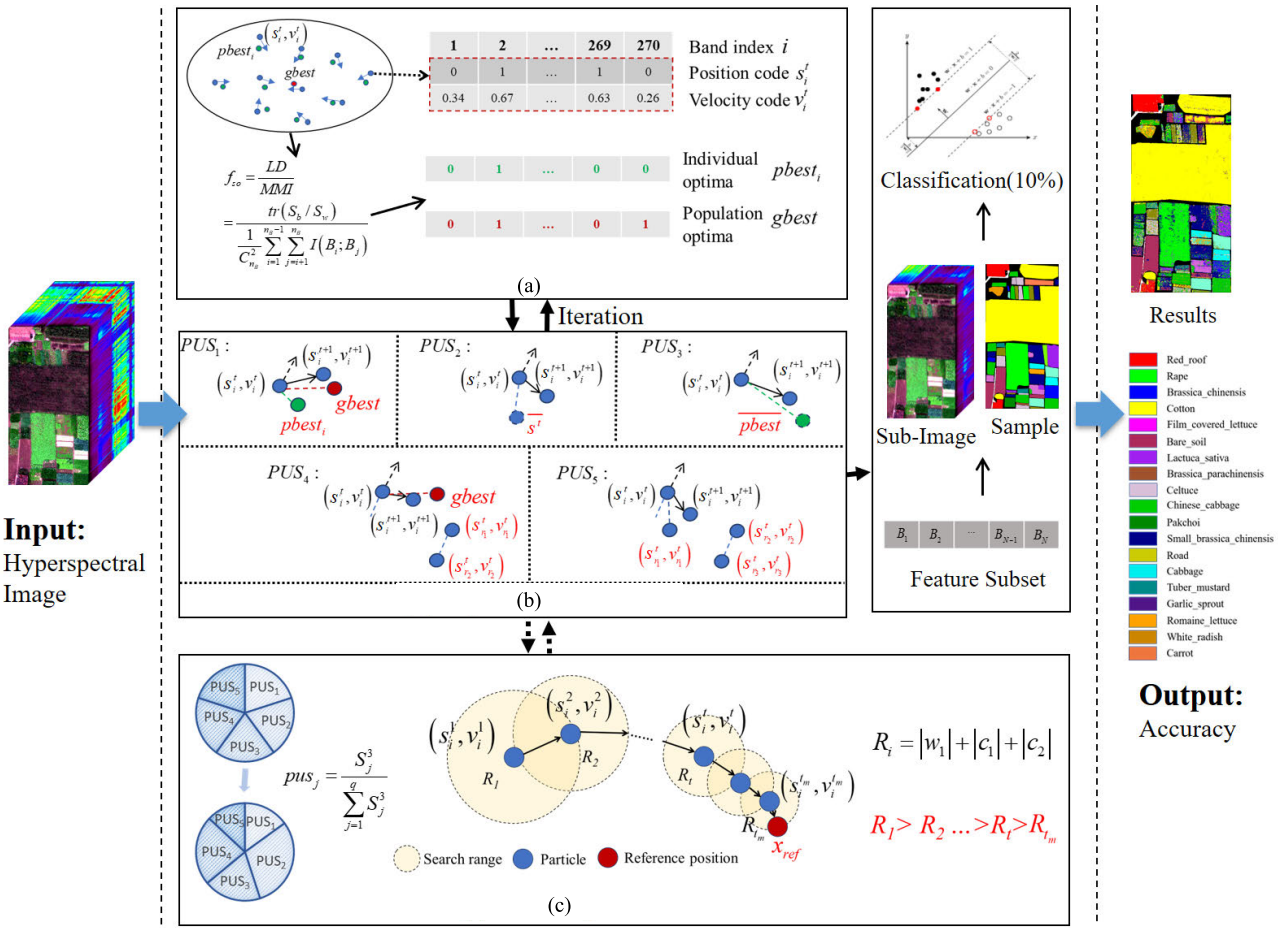


Fig. 2. Main methodological framework of AMSPSO_BS. (a) Encoding and initialization. (b) Population evolution. (c) Self-adaptive mechanism.

$$p(x, y) = \frac{\text{hist}(x, y)}{H \times W} \tag{8}$$

in which $C_{n_B}^2$ represents the combination number when the number of bands is n_B. $I(B_i; B_j)$ relies on the joint probability $p(x, y)$ of the pixel values x and y in the corresponding location between two bands B_i and B_j. Statistically, it is necessary to obtain the 2-D histogram hist(x, y) for calculation, as shown in (8). $H \times W$ refers to the size of the image.

In general, the optimization model for hyperspectral remote sensing image band selection for a solution s in the decisive space Ω is presented in the following equation:

$$\max_{s\in\Omega} f(s) = \max_{s\in\Omega}\left\{\frac{\text{LD}}{\text{MMI}}\right\}. \tag{9}$$

B. Framework of AMSPSO_BS

In this section, an analysis of the proposed AMSPSO_BS method works and a detailed procedure is presented on the basis of the optimization model. The pseudocode is also provided in Algorithm 1.

1) Dataset Preprocess: For different datasets, the pixel values have specific physical meanings, hard for spectrum analysis. Therefore, normalizing the hyperspectral data is an important pretreatment procedure. After this, 10% of samples from each category will be taken to make up the training set while the rest form the test set.

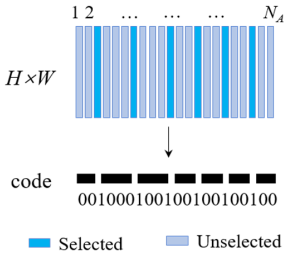


Fig. 3. Initial method for band subset solution.

2) Population Initialization: All bands are randomly disrupted, in which the first n_B terms are selected to form the subset. The binary code of the subset solution serves as the manipulative carrier of the evolutionary operators. And "1" means that the band is chosen while "0" means not. Fig. 3 shows the initialization method, where *ps* individuals can be obtained in this way. All the individuals are then evaluated by (9), in which their current position is viewed as $pbest_i$ and the position of the particle with the highest fitness is considered to be *gbest*.

3) PUS Pool Construction and Population Evolution: In order to provide different meaningful reference positions [34],

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there are five strategies employed in the PUS pool as follows:

$$v_i^t = \begin{cases} wv_i^{t-1} + c_1 r_1\left(pbest_i - s_i^{t-1}\right) \\ \quad + c_2 r_2\left(gbest - s_i^{t-1}\right); & ① \\ wv_i^{t-1} + c_1 r_1\left(\dfrac{\sum_{n=1}^{ps} s_n^{t-1}}{ps} - s_i^{t-1}\right); & ② \\ wv_i^{t-1} + c_1 r_1\left(\dfrac{\sum_{j=1}^{ps} pbest_j}{ps} - s_i^{t-1}\right); & ③ \\ wv_i^{t-1} + c_1 r_1\left(gbest - s_i^{t-1}\right) \\ \quad + c_2 r_2\left(s_{p_1}^{t-1} - s_{p_2}^{t-1}\right); & ④ \\ wv_i^{t-1} + c_1 r_1\left(s_{p_1}^{t-1} - s_i^{t-1}\right) \\ \quad + c_2 r_2\left(s_{p_2}^{t-1} - s_{p_3}^{t-1}\right). & ⑤ \end{cases} \tag{10}$$

$$s_{id}^t = \begin{cases} 1, & \text{Sig mod}\left(v_{id}^t\right) > r_{id} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

where i is the current index of the particle; p_1, p_2, and p_3 refer to the index of the particles randomly selected from the particle swarm; r_1, r_2, and r_{id} are random fractions in the range [0, 1].

The first strategy is the traditional method, representing the guidance of the individual optima and population optima. The second strategy leads the particle through the mean positions of the entire population. The first two strategies affect the update of the particle swarm from the current evolutionary level and the historical optimal level perspectives. The mean optimal positions are adopted in the third approach. The last two strategies are formed through a combination of the main idea of DE and PSO, which takes the influence of other particles into consideration. Therefore, in general, the five strategies in the PUS pool provide different references from different perspectives to avoid rapid convergence.

For each particle i, a randomly selected update strategy j is adopted to adjust its velocity v and position s through (10) and (11), according to the selection probabilities. After the particle i updates, the individual optima $pbest_i$ undergoes fitness growth iteration. And when the entire swarm completes an iteration, the population optima $gbest$ will continue to record the position with the highest fitness.

4) Experience Extraction and Accumulation: For the update of each particle i through strategy j, the change situation of the fitness function is recorded into $nsFlag_{i,j}^{ps\times q}$ and $nfFlag_{i,j}^{ps\times q}$ by (12) and (13). When the current iteration k ends, the record matrix is summed by column to $s_{k,j}^{t_a\times q}$ and $f_{k,j}^{t_a\times q}$ by (14) and (15), respectively. In this way [35], the evolution experience (how to select a strategy for better optimization) is extracted and accumulated with the process of particle motion

$$nsFlag_{i,j}^{ps\times q} = \begin{cases} 1, & f\left(s_i^t\right) > f\left(s_i^{t-1}\right) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

$$nfFlag_{i,j}^{ps\times q} = \begin{cases} 1, & f\left(s_i^t\right) \le f\left(s_i^{t-1}\right) \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

$$s_{k,j}^{t_a\times q} = \sum_{n=1}^{ps} nsFlag_{n,j}^{ps\times q} \tag{14}$$

Algorithm 1 Pseudocode of AMSPSO_BS

Input:
1) *img*: hyperspectral remote sensing image with labels;
2) *ps*: the size of population;
3) t_m: the maximum number of iterations;
4) n_B: the preset size of the selected band subset;
5) t_a: the number of iterations for adjusting the selection probability;
6) w_{max}, w_{min}: the preset extreme value of the inertia weight.

Output:
1) the best solution *gbest*.

initialize a population $\boldsymbol{P_0}$ with $f\left(\boldsymbol{s}_i^0\right)$, $\boldsymbol{s}_i^0 \in \boldsymbol{P_0}$, a PUS pool with q strategies;
$pbest_i \leftarrow \boldsymbol{s}_i^0$, $gbest \leftarrow \boldsymbol{s}_j^0$, $(j = \arg\max_{i=1,2,\ldots,ps}\left\{f\left(\boldsymbol{s}_i^0\right)\right\})$;
$\boldsymbol{nSFlag}^{ps\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{nFFlag}^{ps\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{S}^{ta\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{F}^{ta\times q} \leftarrow \boldsymbol{0}$; $\boldsymbol{pus} \leftarrow \boldsymbol{1/q}$, $k \leftarrow 0$;
for $t \leftarrow 1$ **to** t_m **do**
  for $i \leftarrow 1$ **to** ps **do**
    $\boldsymbol{v}_i^t \leftarrow PUS(\boldsymbol{v}_i^{t-1}, j)$, $\boldsymbol{s}_i^t \leftarrow \text{Bin_Sigmod}\left(\boldsymbol{v}_i^t\right)$ as shown in (10)-(11);
    $pbest_i \leftarrow \boldsymbol{s}_i^n$,$(n = \arg\max_{j=t-1,t}\left\{f\left(\boldsymbol{s}_i^j\right)\right\})$;
    $nsFlag_{i,j}^{ps\times q} \leftarrow f\left(\boldsymbol{s}_i^t\right) > f\left(\boldsymbol{s}_i^{t-1}\right)$, $nfFlag_{i,j}^{ps\times q} \leftarrow f\left(\boldsymbol{s}_i^t\right) \le f\left(\boldsymbol{s}_i^{t-1}\right)$;
  end
  $gbest \leftarrow \boldsymbol{pbest}_i$, $(j = \arg\max_{i=1,2,\ldots,ps+1}\{f(\boldsymbol{pbest}_i) | \boldsymbol{pbest}_{ps+1} = gbest\})$;
  $k \leftarrow k + 1$;
  $S_{k,j}^{t_a\times q} \leftarrow sum\left(nsFlag_{i,j}^{ps\times q}\right)$, $F_{k,j}^{t_a\times q} \leftarrow sum\left(nfFlag_{i,j}^{ps\times q}\right)$, $i = 1, \ldots, ps$;
  if k **equal** t_a **then**
    $\boldsymbol{pus} \leftarrow Update(\boldsymbol{S}^{ta\times q}, \boldsymbol{F}^{ta\times q})$ as shown in (16)–(19), $k \leftarrow 0$;
  end
  $w \leftarrow w_{\max} - \frac{t}{t_m}(w_{\max} - w_{\min})$, $c \leftarrow \frac{(w+1+2\sqrt{w})}{2}$;
  $\boldsymbol{nSFlag}^{ps\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{nFFlag}^{ps\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{S}^{ta\times q} \leftarrow \boldsymbol{0}$, $\boldsymbol{F}^{ta\times q} \leftarrow \boldsymbol{0}$;
end
Return the solution *gbest*.

$$f_{k,j}^{t_a\times q} = \sum_{n=1}^{ps} nfFlag_{n,j}^{ps\times q}. \tag{15}$$

5) Self-Adjustive Mechanism: When the matrices $\boldsymbol{S}^{ta\times q}$ and $\boldsymbol{F}^{ta\times q}$ are full, i.e., the number of iterations reaches the preset t_a, the selection probability is updated by (16)–(19). It is apparent that the more evolutions and fewer degradations the strategy leads to, the higher the selection probability to be assigned

$$S_j^1 = \sum_{n=1}^{t_a} s_{n,j}^{t_a\times q} \tag{16}$$

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$$S_j^2 = \begin{cases} \varepsilon, & S_j^1 = 0 \\ S_j^1, & \text{otherwise} \end{cases} \tag{17}$$

$$S_j^3 = \frac{S_j^1}{S_j^2 + \sum_{n=1}^{t_a} f_{n,j}^{t_a \times q}} \tag{18}$$

$$\text{pus}_j = \frac{S_j^3}{\sum_{j=1}^{q} S_j^3} \tag{19}$$

where q is the number of strategies involved in AMSPSO_BS ($q = 5$), pus_j refers to the selection probability of the jth strategy, and ε represents a value close to 0.

In addition, the parameters for the particle motion are also adjusted as the iteration progresses by (20) and (21). Linearly reduced motion parameters facilitate the transition from global to local search capabilities [36]

$$w = w_{\max} - \frac{t}{t_m}(w_{\max} - w_{\min}) \tag{20}$$

$$c_1 = c_2 = \frac{\left(w + 1 + 2\sqrt{w}\right)}{2} \tag{21}$$

where $w_{\max}$ and $w_{\min}$ are the preset extreme values of the inertia weight.

6) Stopping Condition: The related operations for the population in steps 3)–5) are repeated t_m times.

IV. Experiments and Results

In order to verify the effectiveness and generalization ability of AMSPSO_BS, experiments on four hyperspectral remote sensing image datasets have been performed. The proposed method was compared with some non-EC-based methods and four typical EC-based methods under the same metric of LD/MMI. The accuracy of the original hyperspectral image with full bands was also taken as a reference.

A. Experimental Description

The proposed AMSPSO_BS method was compared with other band selection methods as follows: the typical non-EC-based methods, including maximum variance principal components analysis (MVPCA) [37], MI-based [38], unsupervised hyperspectral band selection by dominant set extraction (DSEBS) [39], sparse representation based band selection (SpaBS) [40], end-to-end framework for band selection (BS-Nets) [41], and EC-based methods with the same metric (LD/MMI), i.e., a GA-based method [42], a DE-based [43], a PSO-based method [44], and a sine cosine algorithm (SCA)-based method [45]. The critical parameters in the above EC-based methods are the same as the references and listed in Table I. It is worth noting that the population size and the maximum number of iterations are the same. And the solution evaluation is conducted once for each individual per iteration, ensuring that all the EC-methods consume the same number of solution evaluations.

In addition, since the solutions of these methods are band subsets, which are hard to compare directly, the classification performance of their corresponding multiband subimages served as the evaluation criteria, with the linear SVM classifier and the k-nearest neighbors (KNN) classifier used. The penalty coefficient C and the kernel parameter σ of SVM were set to 1 and 1, respectively. The number of nearest neighbors of KNN was set to 8. For each category, 10% of the labeled samples were selected and used to train the classifiers, and the remaining were preserved for assessment. Moreover, the overall accuracy (OA) and the Kappa coefficient (Kappa) obtained were utilized as the comparison metrics.

TABLE I
Key Parameters of the Tested EC Methods

Size of population ps			50
Maximum number of iterations t_m			100
EC methods	GA-based	p_{cross}	0.7
		p_{muta}	0.02
	DE-based	F	0.25
		C	0.6
	PSO-based	w	0.7298
		c_1, c_2	1.496, 1.496
	SCA-based	a	1

Notes: p_{cross}: crossover probability. p_{muta}: mutation probability. F: factor of mutation. C: factor of crossover. w: inertia weight. c_1, c_2: acceleration factor. a: amplitude of the parameter r_1.

B. Experiment 1: Indian Pines Dataset

The Indian Pines hyperspectral remote sensing image dataset [46] captured the Indian Pines region in June 1992 by the airborne visible infrared imaging spectrometer (AVIRIS) sensor. This public image dataset is composed of 200 bands ranging from 400 to 2500 nm after removing some channels disturbed by serious noise and water absorption. The image size is 145 × 145 pixels at a spatial resolution of 20 m. There are a large number of mixed pixels, which can be divided into 16 agricultural classes in total. Fig. 4(a) and (b) show the hyperspectral image data cube and the ground truth. The average spectral curves for all the classes are shown in Fig. 4(c), so as to exhibit the radiation differences.

For the band selection task, the size of the subset, which is difficult to determine in advance, affects the application performance of the solutions greatly. Thus, the qualitative comparison was performed on the classification accuracy curve, with the preset size of the subset changed from 3 to 60.

Fig. 5 presented the results of the whole tested methods in the Indian Pines dataset. When the preset size of the selected band subset is small, the subset optimized by AMSPSO_BS shows better classification performance than the subsets obtained by the other methods. On the SVM classifier, the corresponding curve of AMSPSO_BS is located above the curves of others. In addition, the multiband subimage corresponding to the selected bands performs better than the original hyperspectral remote sensing image, as a result of removing the noisy and redundant bands. On the KNN classifier, AMSPSO_BS does not achieve the best performance but is superior to all the EC-based methods. And only MI reaches the classification accuracy of the original hyperspectral image.

For an intuitive and quantitative comparison, Table II lists the mean and the standard deviation of classification accuracy for all the tested methods, which can reflect the comprehensive

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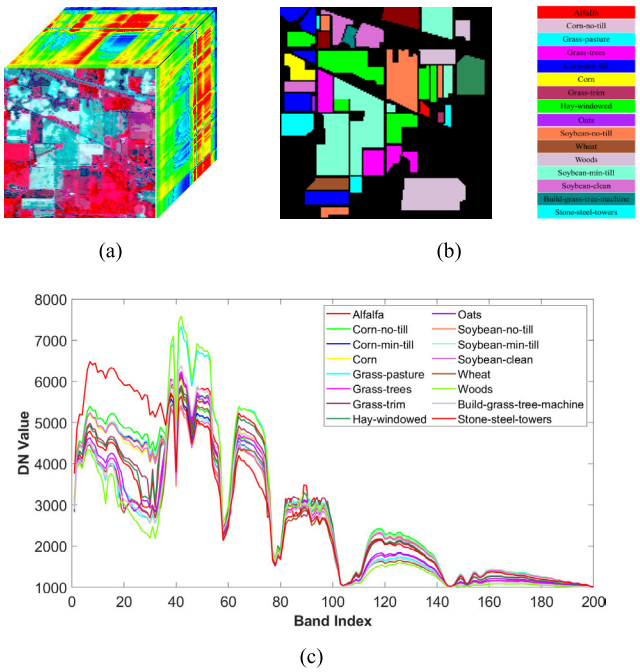


Fig. 4. Indian Pines dataset. (a) Image cube. (b) GT with legends. (c) Spectral curves.

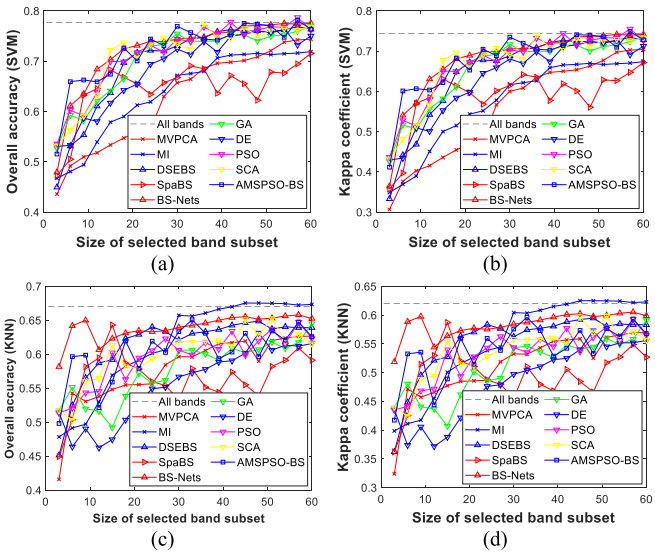


Fig. 5. Classification accuracy curves on the Indian Pines dataset. (a) OA-SVM. (b) Kappa-SVM. (c) OA-KNN. (d) Kappa-KNN.

extraction capabilities of critical information. On the SVM classifier, the proposed AMSPSO_BS method shows significant advantages, as expected. And the EC-based methods perform better than traditional methods. On the KNN classifier, the subset solution of BS-Nets reaches the highest accuracy. But all the EC-based methods perform badly, suggesting that the fitness function (LD/MMI) could hardly realize decisive spectrum selection of hyperspectral datasets with noise suitable for the KNN classifier.

The OA and Kappa can evaluate the general classification performance. But it can hardly reflect the details of different categories with various distribution and sizes. Therefore, producer accuracy (PA) was utilized for a comprehensive comparison. There are significant differences in the results of different methods according to the classification curves when the preset band size is 6 with the SVM classifier used. So, the tests are conducted on the condition and the results are presented in Table III.

TABLE II

CLASSIFICATION ACCURACY ON THE INDIAN PINES DATASET (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Method	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
All bands	77.72	74.44	67.08	62.00
MVPCA	62.75±9.73	55.76±12.62	57.83±5.06	51.22±5.86
MI	63.82±8.63	57.29±10.98	**61.53±6.94**	**55.53±8.07**
DSEBS	69.78±9.21	64.74±11.37	61.32±4.92	55.26±5.81
SpaBS	64.75±5.93	58.74±7.76	56.51±4.07	49.55±4.76
BS-Nets	**72.19±7.29**	**67.64±9.20**	**63.99±1.88**	**58.43±2.18**
GA	70.30±7.14	65.42±8.89	57.70±4.26	50.95±5.11
DE	68.54±7.75	63.21±9.78	55.59±5.47	48.40±6.54
PSO	71.59±7.00	66.89±8.75	59.60±3.83	53.18±4.56
SCA	71.31±7.28	66.57±9.11	60.00±3.93	53.68±4.61
proposed	**72.26±6.34**	**67.78±7.92**	60.33±3.93	54.08±4.62

It could be found that the proposed method achieves relatively balanced classification performance for all categories to reach the highest OA in the tested methods. Among the 16 categories, it performed the best in seven of them, including alfalfa, corn-no-till, corn-min-till, corn, soybean-no-till, wheat, and build-grass-tree-machine. For the oats with the smallest sample size, all methods cannot effectively distinguish. For the soybean-min-till with the largest sample size, MVPCA utilizes the variance for band selection and achieves optimal classification. Besides, for the soybean-min-till and woods, better performance is achieved by the whole methods than all bands, suggesting the subset results obtained are more discriminant for the two categories.

C. Experiment 2: WHU-Hi Datasets

The three WHU-Hi datasets [47] were acquired in the agricultural farming areas in Hubei province, China. Since all the datasets were captured by a Headwall Nano-Hyperspec sensor mounted on an unmanned aerial vehicle platform, they have the same spectral range of 400–1000 nm.

The first dataset is the WHU-Hi-LongKou dataset, which covers a simple agricultural scene in Longkou Town with six crop species and three farmland classes. The spatial resolution of the dataset is approximately 0.463 m, and 550 × 400 pixels are recorded in a single band.

The second one is the WHU-Hi-HanChuan dataset, which captures a rural-urban fringe zone with 16 categories containing artificial buildings, water, and various crops. The size of a single band is 1217 × 303 pixels, and the resolution reaches about 0.109 m. There is shadow occlusion in the dataset, which may affect the interpretation accuracy. Compared with the first dataset, WHU-Hi-HanChuan is harder to classify with the ground objects distributed in strips.

The last dataset is the WHU-Hi-HongHu dataset. This dataset acquired in the city of Honghu covers a complex agricultural scene with different cultivars of the same crop.

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TABLE III
PA FOR DIFFERENT CATEGORIES ON THE SVM CLASSIFIER WITH THE PRESET SIZE OF 6 (%)

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16
All bands	70.73	74.32	55.82	64.79	85.48	94.37	72.00	99.77	44.44	67.73	77.86	66.23	99.46	93.50	59.08	74.70
MVPCA	0	0	0	0	48.85	91.02	0	70.00	0	0	**98.01**	0	53.80	95.25	4.03	74.70
MI	0	0	3.08	0	0	89.95	0	91.86	0	0	94.98	**29.64**	17.93	97.54	0.58	26.51
DSEBS	7.32	38.91	4.15	0	11.52	77.93	0	76.51	0	21.17	87.01	3.94	88.04	95.61	10.66	72.99
SpaBS	0	32.68	0	0	10.37	79.60	0	53.26	0	0	94.16	1.13	88.59	96.05	12.68	65.06
BS-Nets	0	32.14	35.54	14.08	**75.58**	**92.24**	12.00	95.81	0	28.15	87.64	3.00	98.37	95.52	18.16	72.29
GA	14.63	42.26	23.96	8.45	61.75	80.97	8.00	97.67	0	0.11	88.50	16.32	96.20	96.40	34.01	**75.90**
DE	0	35.64	0	2.82	13.59	77.63	0	95.35	0	0	93.71	1.50	86.96	97.28	18.44	73.49
PSO	0	45.21	0	0	64.29	83.26	**16.00**	**99.77**	0	36.73	89.14	0	92.93	95.52	31.41	69.88
SCA	0	42.80	0.54	9.39	5.07	86.30	8.00	99.30	0	45.88	83.93	0.19	75.00	**97.80**	10.09	65.06
proposed	**56.10**	**52.14**	**36.81**	**18.78**	55.33	86.61	12.00	97.44	0	**50.00**	84.16	15.95	**98.37**	96.40	**34.58**	71.08

Notes: C1. Alfalfa, C2. Corn-no-till, C3. Corn-min-till, C4. Corn, C5. Grass-pasture, C6. Grass-trees, C7. Grass-trim, C8. Hay-windowed, C9. Oats, C10. Soybean-no-till, C11. Soybean-min-till, C12. Soybean-clean, C13. Wheat, C14. Woods, C15. Build-grass-tree-machine, C16. Stone-steel-towers.

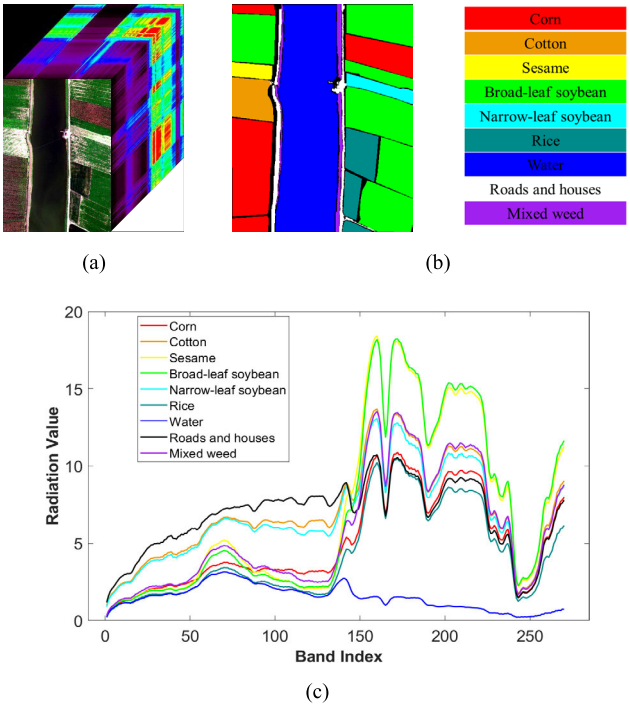


Fig. 6. WHU-Hi-LongKou dataset. (a) Image cube. (b) GT with legends. (c) Spectral curves.

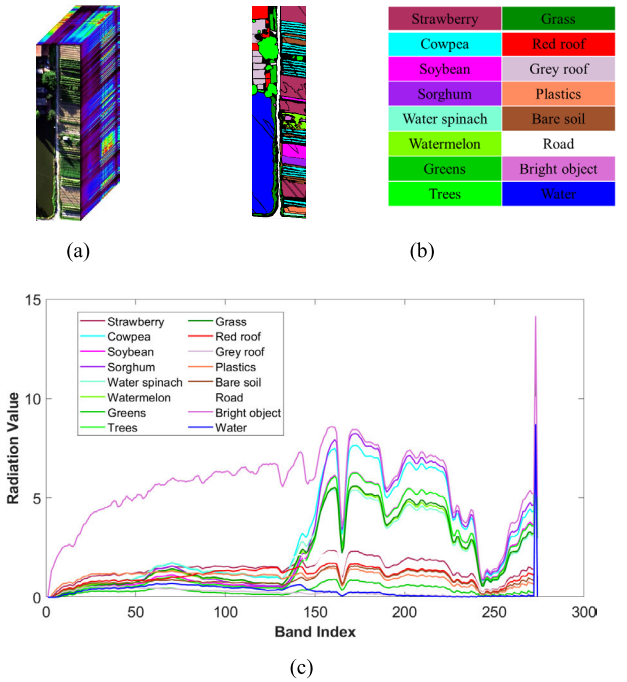


Fig. 7. WHU-Hi-HanChuan dataset. (a) Image cube. (b) GT with legends. (c) Spectral curves.

Their spectral curves are much more similar, so that the fine classification is more difficult. The size of the imagery is 940 × 475 pixels, at the highest spatial resolution of about 0.043 m. Different from the WHU-Hi-HanChuan dataset, most objects in the image are blocky.

The WHU-Hi datasets are all hyperspectral images with a high spatial resolution, which are known as H^2 imagery. For this type of imagery, more precise identification of ground objects can be achieved, supporting more refined applications. The details of the datasets are presented in Figs. 6–8.

All the tested methods (MVPCA, MI, DSEBS, SpaBS, BS-Nets, GA-based, DE-based, PSO-based, SCA-based, and AMSPSO_BS) were utilized to obtain the best band subset in each WHU-Hi dataset. The multiband subimage corresponding to the solution was then classified and compared with the original hyperspectral remote sensing image.

Fig. 9 and Tables IV–VI make qualitative and quantitative comparisons between different band selection methods. Differing from the Indian Pines dataset, the EC-based methods can select band subsets with better application performance than the conventional method, meaning that redundancy removal is the sole objective of band selection for the WHU-Hi hyperspectral remote sensing image datasets.

As shown in Fig. 9, the accuracy curves tend to be smooth when the preset size n_B continues to increase. In addition, the maximum of OA and Kappa delivered by the band selection methods still do not reach the classification accuracy of the original hyperspectral image on SVM. This may suggest that 10% of the training samples fail to exhibit a significant Hughes phenomenon. To fully consider the effects of the band selection methods, a small sample analysis was required, which is described in Section V.

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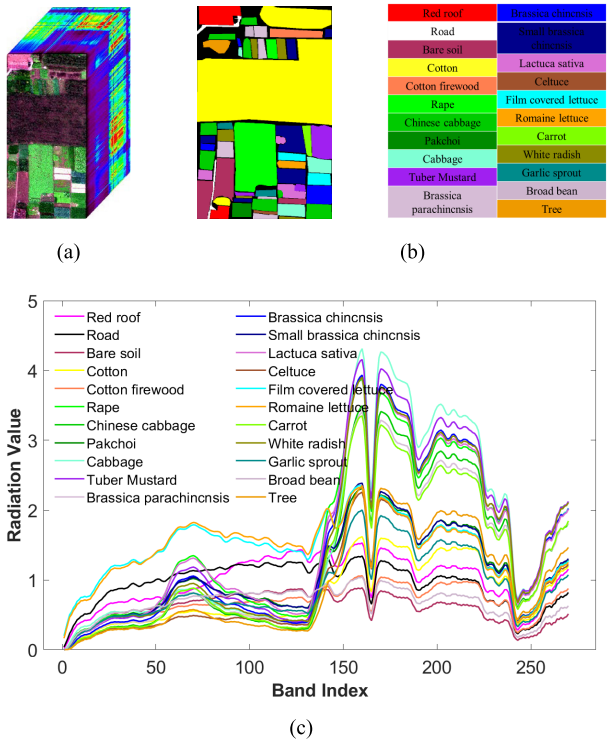


Fig. 8. WHU-Hi-HongHu dataset. (a) Image cube. (b) GT with legends. (c) Spectral curves.

TABLE IV

CLASSIFICATION ACCURACY ON THE WHU-HI-LONGKOU DATASET (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Method	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
All bands	98.85	98.49	94.28	92.43
MVPCA	91.37±2.41	88.51±3.31	88.86±1.52	85.13±2.05
MI	91.57±2.18	88.80±2.94	90.69±1.73	87.65±2.31
DSEBS	95.67±1.75	94.28±2.33	93.54±0.72	91.46±0.95
SpaBS	93.33±7.03	91.08±9.71	91.96±4.17	89.32±5.65
BS-Nets	96.36±3.07	95.19±4.11	93.66±1.48	91.61±1.96
GA	96.55±2.68	95.44±3.59	**94.28±1.67**	**92.44±2.23**
DE	96.71±1.46	95.66±1.94	93.60±0.42	91.54±0.56
PSO	**96.76±1.63**	**95.72±2.17**	94.22±0.56	92.36±0.75
SCA	96.69±1.67	95.63±2.22	93.95±0.61	92.00±0.81
proposed	**96.91±1.55**	**95.93±2.06**	**94.35±0.62**	**92.54±0.83**

For the WHU-Hi-LongKou dataset, most of the tested band selection methods can optimize the problem well and achieve nearly saturated performance on the two classifiers. It is therefore difficult to distinguish the best method from the others, except for MVPCA, MI, and SpaBS. From Table IV, the PSO-based method shows a slight advantage over the other methods in the average accuracy, and the AMSPSO_BS achieves the highest accuracy. And on the KNN classifier, the performance of most methods is better than that of all bands. Overall, subset evaluation on the WHU-Hi-LongKou dataset should be deepened with a small sample analysis to confirm the effectiveness of the proposed method.

As for the WHU-Hi-HanChuan dataset, the accuracy curve of AMSPSO_BS is located above the curves of the other methods on the SVM classifier, consistent with the results listed

TABLE V

CLASSIFICATION ACCURACY ON THE WHU-HI-HANCHUAN DATASET (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Method	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
All bands	86.30	83.94	78.04	74.08
MVPCA	54.48±1.49	43.99±2.18	54.15±2.93	45.05±3.50
MI	56.59±5.11	46.47±6.93	56.37±6.02	47.68±7.51
DSEBS	76.36±7.97	71.86±10.44	71.01±9.94	65.72±11.68
SpaBS	72.96±7.13	67.77±9.41	72.05±6.78	66.85±8.49
BS-Nets	71.97±8.09	66.55±10.08	**72.94±5.66**	**68.23±6.76**
GA	77.29±8.93	72.93±11.62	72.32±10.46	67.20±12.63
DE	76.70±9.40	72.12±12.23	67.83±10.26	61.86±12.12
PSO	76.96±9.41	72.42±12.71	72.55±8.84	67.52±10.60
SCA	**77.65±6.47**	**73.52±8.11**	72.61±7.66	67.70±8.78
proposed	**77.68±8.88**	**73.30±12.28**	**73.22±7.75**	**68.29±9.48**

TABLE VI

CLASSIFICATION ACCURACY ON THE WHU-HI-HONGHU DATASET (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Method	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
All bands	84.66	80.49	80.98	75.30
MVPCA	65.90±3.56	54.37±5.74	64.33±3.48	52.16±5.24
MI	69.95±1.13	60.75±1.54	69.82±0.87	60.50±1.13
DSEBS	79.03±6.03	72.77±8.68	77.35±5.43	70.39±7.59
SpaBS	77.22±5.63	70.32±8.47	76.11±5.30	68.74±7.56
BS-Nets	80.91±3.23	75.49±4.38	**78.84±2.40**	**72.57±3.21**
GA	**81.10±3.74**	**75.72±5.04**	78.35±2.36	71.75±3.07
DE	80.76±4.57	75.23±6.31	78.26±3.24	71.62±4.41
PSO	80.93±4.65	75.44±6.48	78.67±3.55	72.20±4.87
SCA	81.01±4.49	75.56±6.23	78.72±3.31	72.29±4.50
proposed	**81.20±4.51**	**75.81±6.24**	**78.91±3.44**	**72.53±4.70**

in Table V. When the preset size n_B is small, the proposed method can obtain solutions that are suitable for distinguishing between the different classes. On the SVM classifier, EC-based methods are superior to non-EC-based methods at all subset sizes. On the KNN classifier, the performance is closer for most methods. And the performance of BS-Nets is better than all bands when the size increases.

For the WHU-Hi-HongHu dataset, the curves of the EC-based methods and BS-Nets are superior to that of the other methods. In addition, the performance of the final solutions optimized by the EC-based methods shows no great differences in either the SVM classifier or the KNN classifier. That suggests that the metric LD/MMI with a strong category separability would help to obtain a band subset with superior performance. And according to Table VI, AMSPSO performs the best among them.

Therefore, in brief, band selection tasks can be difficult to optimize when faced with H^2 datasets with abundant spectral information and detailed spatial distribution information. But the proposed method still has advantages over the traditional methods with the proper metric.

V. DISCUSSION AND ANALYSES

Most experiments confirmed the great optimization ability of the proposed method. However, more discussions and analyses are required from different perspectives.

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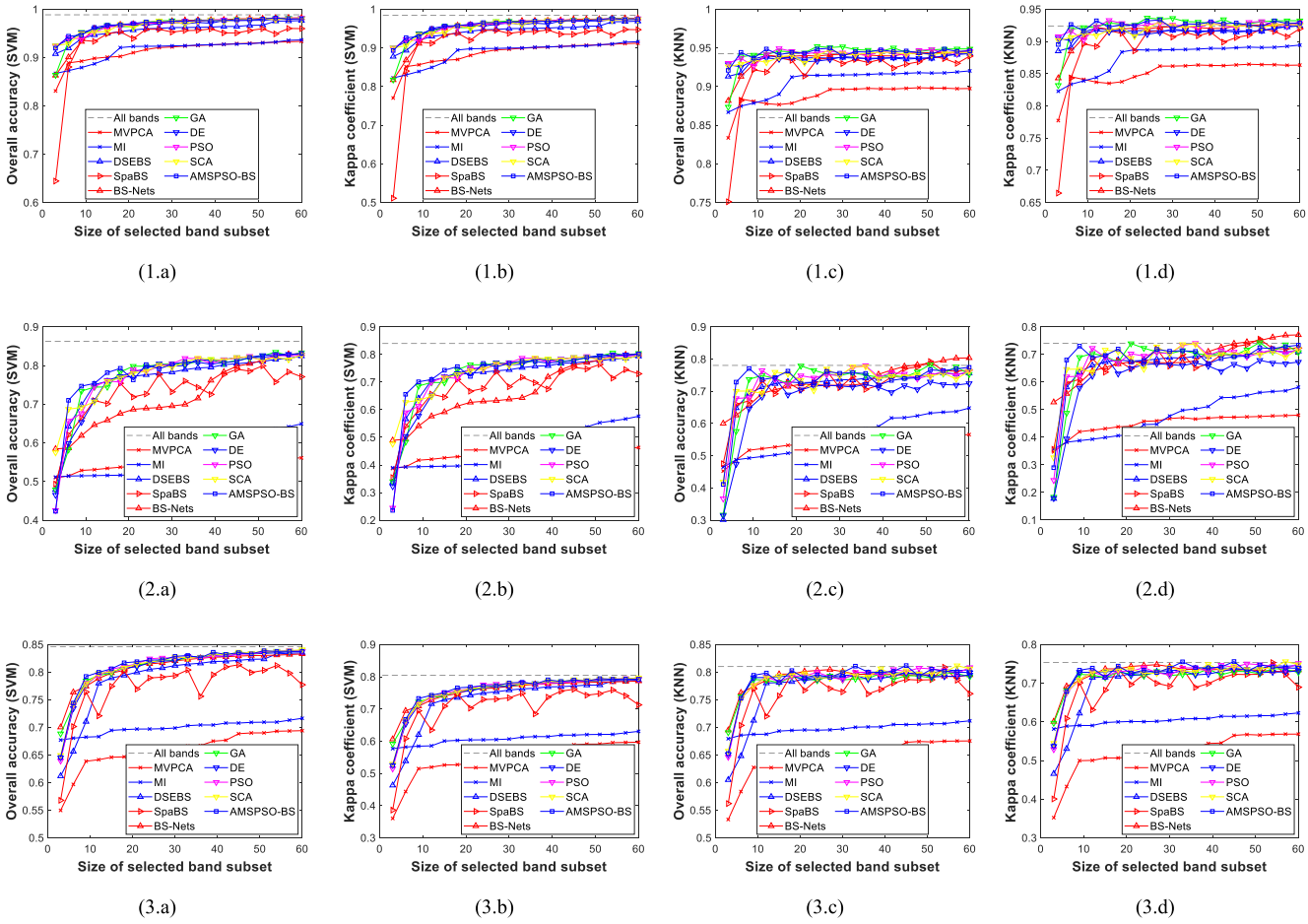


Fig. 9. Classification accuracy curves. (1) WHU-Hi-LongKou dataset. (2) WHU-Hi-HanChuan dataset. (3) WHU-Hi-HongHu dataset. (a) OA-SVM. (b) Kappa-SVM. (c) OA-KNN. (d) Kappa-KNN.

TABLE VII

CLASSIFICATION ACCURACY WITH DIFFERENT METRICS (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Metric	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
Acc	74.24±6.79	70.10±8.43	69.10±1.64	64.38±1.89
Cor	69.81±7.59	64.80±9.52	60.18±3.84	54.31±4.43
OIF	69.27±7.41	64.10±9.34	59.92±3.64	53.61±4.25
LD/MMI	72.26±6.34	67.78±7.92	60.33±3.93	54.08±4.62

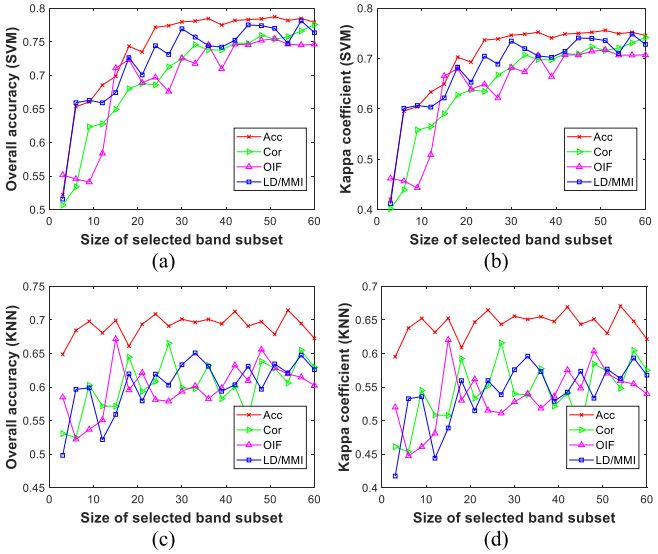


Fig. 10. Classification accuracy curves with different metrics. (a) OA-SVM. (b) Kappa-SVM. (c) OA-KNN. (d) Kappa-KNN.

A. Ablation Study Analysis

1) Analysis of Fitness Function: The fitness function (LD/MMI) determines "what to select," which remains the same in the previous experiments. To verify the effectiveness of the fitness function, we took the Indian Pines dataset as an example to compared with other typical evaluation functions, including the accuracy and the percentage of eliminated bands (Acc) used in [9], correlation-based (Cor) selection in [43], and optimal index factor (OIF) in [48]. The classification performance is recorded in Fig. 10 and Table VII.

It can be found that subset solutions obtained by Acc perform the best on both classifiers. But in fact, Acc is a kind of wrapper-based metric with higher computational complexity. For the rest, the classification curves of the proposed metric are above the others on the SVM classifier. While on KNN, the curves are mixed, hard to evaluate. From Table VII, the proposed metric is proved to be an effective fitness function in the filter-based methods.

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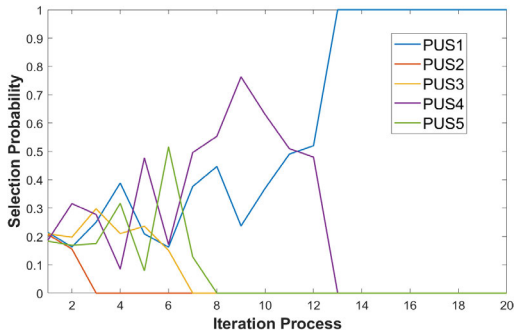


Fig. 11. Change curves of the selection probability of different PUSs.

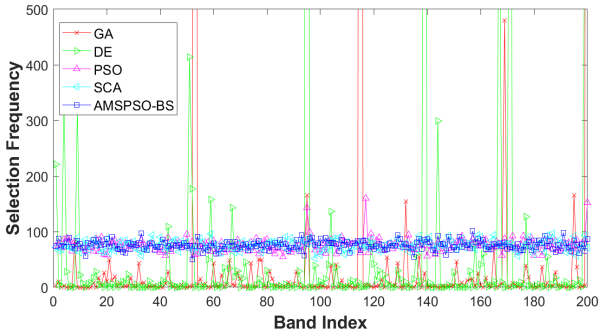


Fig. 12. Frequency histograms of selection for different EC-based methods.

2) Analysis of the Optimization Method: An adaptive multistrategy PSO was designed to solve the problem of "how to select." Some analyses were conducted in the proposed method to study the optimization process for validation of the superiority.

On the one hand, the change curves of the selection probability of all the PUSs were drawn in Fig. 11. From the figure, the selection probability of different PUSs will fluctuate. The PUS which promotes particle update will dominate in the PUS pool gradually. In the early stage, all the strategies participate in the particle update and help with global exploration while in the later, the first PUS is selected for local exploitation. It suggests that the self-adaptive mechanism of the PUS pool can dynamically adjust the approach of particle motion based on the progress of the optimization process, which helps to find a better band subset.

On the other hand, the frequency histograms of selection for each band were recorded during the optimization process as shown in Fig. 12. Compared with other EC-based methods, AMSPSO_BS achieves more even distribution of selection for each band, indicating a more sufficient exploration of the decisive space for best subset solutions.

B. Small Sample Analysis

As can be seen from the results on the WHU-Hi-LongKou dataset described in the last section, all the solutions obtained by different methods achieved nearly saturated performance. Thus, it is necessary to analyze the small sample classification further. Driven by this consideration, fewer samples (100 for each category) from the WHU-Hi-LongKou dataset were selected to train the classifiers.

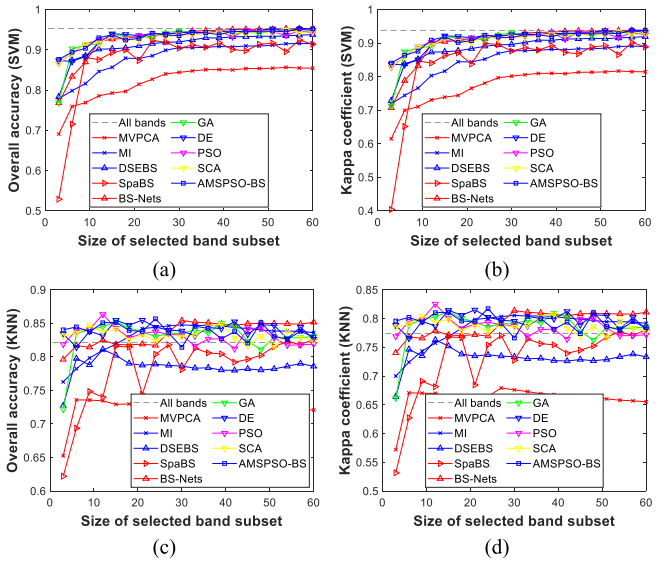


Fig. 13. Small sample analysis on WHU-Hi-LongKou. (a) OA-SVM. (b) Kappa-SVM. (c) OA-KNN. (d) Kappa-KNN.

TABLE VIII
SMALL SAMPLE ANALYSIS ON WHU-HI-LONGKOU DATASET (%) (THE MEAN OA/KAPPA ± THE STANDARD DEVIATION)

Method	Classifier: SVM		Classifier: KNN	
	M_OA±Std	M_K±Std	M_OA±Std	M_K±Std
All bands	95.27	93.84	82.09	77.44
MVPCA	82.21±4.42	77.50±5.37	72.68±1.83	66.15±2.19
MI	88.22±4.17	84.86±5.22	83.00±2.38	78.34±2.98
DSEBS	91.18±3.51	88.64±4.35	78.50±1.57	73.20±1.84
SpaBS	87.54±9.30	84.09±11.59	78.39±5.15	73.05±6.16
BS-Nets	92.46±4.76	90.28±5.98	83.56±1.78	79.13±2.19
GA	92.81±3.96	84.94±4.86	82.99±2.71	**80.79±3.12**
DE	**93.24±2.55**	**91.24±3.24**	**83.97±1.04**	79.65±1.26
PSO	93.12±2.01	91.07±2.58	83.11±1.25	78.61±1.51
SCA	93.05±2.03	90.99±2.59	83.23±0.92	**80.79±1.09**
proposed	**93.30±2.00**	**91.31±2.25**	**83.78±1.09**	79.43±1.31

The results are provided in Fig. 13 and Table VIII. In the figure, due to the insufficient classifier training caused by the limited samples, the accuracy of some methods fluctuates dramatically. Most methods have achieved similar or better classification performance compared with all bands, especially on the KNN classifier, indicating that the selected bands are discriminant. From Table VIII, the subset solutions obtained by AMSPSO_BS achieve the best application performance on the SVM classifier. And it obtains suboptimal results on KNN.

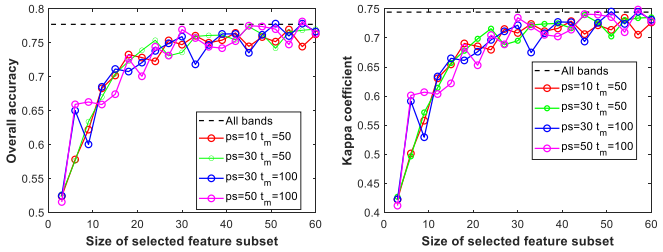
C. Parameter Sensitivity Analysis

The experimental results in Section IV confirmed that the solution optimized by AMSPSO_BS shows better classification performance. Nevertheless, there are two critical parameters in AMSPSO_BS that need to be preset properly. They are the preset size of the population (ps) and the maximum number of iterations (t_m). The ps and t_m parameters mainly control the breadth and depth of the search process, which both affect the optimization efficiency. In this section, the effects of these parameters were analyzed through the corresponding sensitivity analysis on the Indian Pines dataset.

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TABLE IX
CLASSIFICATION ACCURACY ON DIFFERENT PARAMETER SETTINGS (%)

ps	t_m	OA		Kappa	
		Mean±Std	Max	Mean±Std	Max
10	50	71.77±6.71	76.76	67.18±8.37	73.48
30	50	72.07±6.81	77.38	67.50±8.50	73.93
30	100	72.09±6.42	77.83	67.57±8.03	74.53
50	100	72.26±6.34	78.14	67.78±7.92	74.88



(a) (b)

Fig. 14. Parameter sensitivity analysis on Indian Pines. (a) OA. (b) Kappa.

The two parameters were set to {10, 30, 30, 50} and {50, 50, 100, 100} in order. Comparisons are provided using the change curves, the mean, and the maximum of OA and Kappa on the SVM classifier, which are shown in Fig. 14 and Table IX.

It can be observed from Fig. 14 that there are random undulations in the accuracy curves obtained with the different parameter settings, which cannot reveal any regularity. Fortunately, from Table IX, we can conclude that the change of t_m has a distinct impact on the maximum accuracy when ps remains. The first two rows and the last two rows show that ps affects both the mean and the maximum accuracy. This is reasonable since the larger the size of the population, the more regions will be explored in the optimization. So, these parameters are supposed to have higher values to make sure the EC-based method convergence.

D. Subset Component Analysis

The band subsets are the direct results of all the methods, so it is meaningful to study the components of the band subsets. It should reveal the critical bands suitable for classification, which is beneficial to the design of the characteristic spectrum. The best subset solutions obtained under the smallest band subset size 3 and different preset sizes from 12 to 60 with a step size of 12 for the Indian Pines dataset and the three WHU-Hi datasets are shown in Fig. 15. Moreover, the subset solutions obtained by different band selection methods are compared for comprehensive validation of the effectiveness. Indian Pines dataset was taken as an example to conduct some discussions. For better visualization, the band composition results with large accuracy differences and small subset sizes are drawn in Fig. 16 as well. Specifically, the subset size n_B is set to 6.

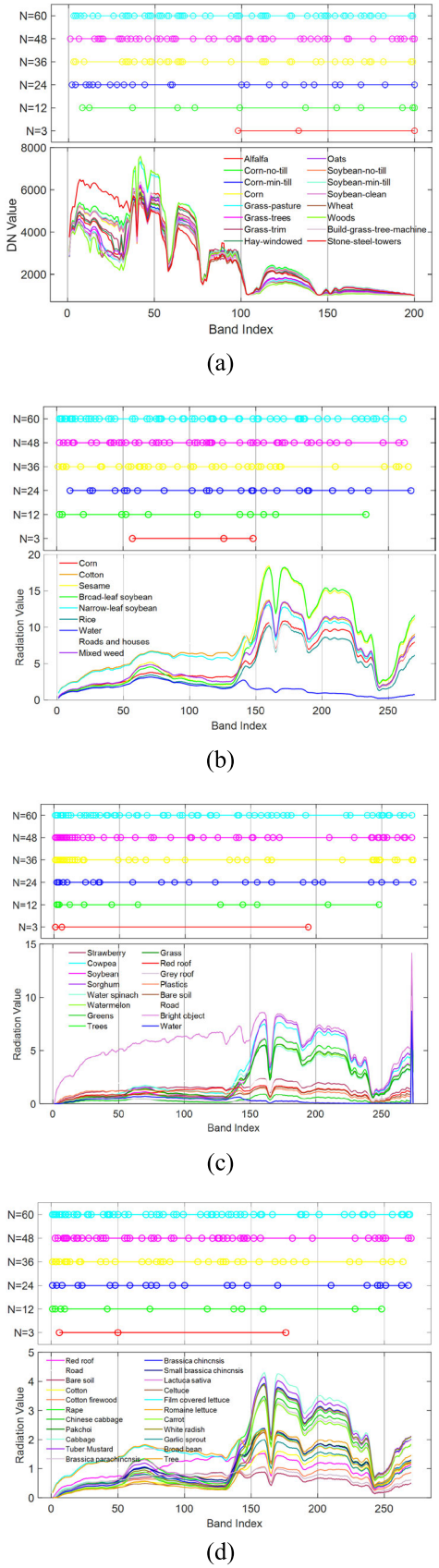


Fig. 15. Subset components analyses on different datasets. (a) Indian Pines dataset. (b) WHU-Hi-LongKou dataset. (c) WHU-Hi-HanChuan dataset. (d) WHU-Hi-HongHu dataset.

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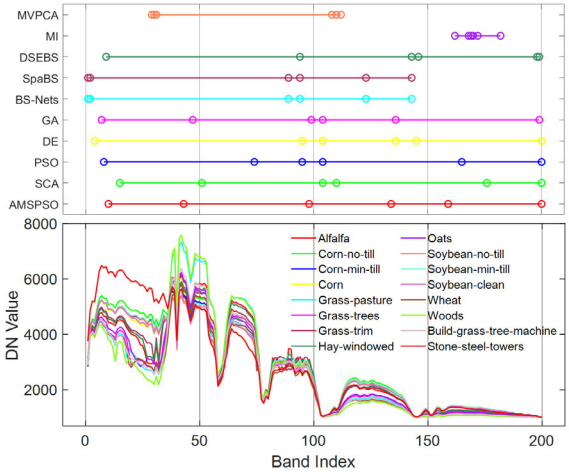


Fig. 16. Subset components analyses for different methods on Indian Pines.

As mentioned above, the spectral range of the Indian Pines dataset has a lower resolution than the other datasets. The differences between the classes are mainly concentrated on the first 80 bands, including the visible and near-infrared bands. Therefore, the majority of the selected bands are located in this spectral range. The bands in the peak from 80 to 100 are rarely chosen as the components of the subset solutions. Visual interpretation can also allow the conclusion that there is serious intersection between the curves of the different categories, which is more likely to cause inaccurate classification. In the last 100 bands, a small number of bands are selected, which are mainly distributed in the upper and lower edges of the peaks.

In the following, we analyze the subset components of the WHU-Hi dataset. As for the WHU-Hi-LongKou dataset, the distribution of the subset components is even. The peaks of the spectral curves are seldom sampled, and the bands of the upper and lower edges are preferred.

From the results of Fig. 15 (3), most of the selected bands come from the first 150 bands. The curves in this range are much gentler, but have lower redundancy, so the bands in this range are more likely to be chosen. In contrast, the peak after band 150 is ignored, which is unreasonable, due to the better category separability of these bands.

For the WHU-Hi-HongHu dataset, it can be found that the curves are relatively stable in the first 150 spectral bands (visible spectrum), which has a high proportion of the components in the subset, especially the first 50 bands. The low redundancy of these bands may account for their selection. In bands 170–220 (near-infrared spectrum), the number of selected bands is small. That is because the information redundancy in these bands is high. Around band 250, the curve changes dramatically and more bands are selected.

As for the comparison of different methods in Fig. 16, the subset solutions obtained by MVPCA and MI present obvious aggregation, resulting in more redundancy. Most methods select spectrum near band 10 and band 90. And the results of all EC-based methods cover the whole spectrum. Among them, the distribution of the proposed method is the most uniform and precisely covers the peak area. That may account for the highest classification accuracy.

In general, the EC-based band selection methods with the only metric of LD/MMI can obtain subsets with low redundancy from all the bands. However, whether these selected bands are suitable for classification needs to debate.

E. Computational Complexity Analysis

We can assume that there are N_A bands and $n_P = H \times W$ pixels in the hyperspectral remote-sensing image. Thus, the computational complexity of the proposed method can be concluded as follows.

1) Population Initialization: There are ps individuals in the particle swarm and each particle has N_A dimensions. So, the computational complexity is $O(ps \times N_A)$.

2) Solution Evaluations and Reference Update: To evaluate the multiband subimage made of n_B bands and K categories, the computational complexity for the fitness function f is $O_1 = O(ps \times (K + C_{n_B}^2 \times n_P))$. For the update of the reference optima, comparisons are conducted between the individuals and references. The operations need computation consumption of $O_2 = O(ps)$.

3) Main Loop of Adaptive Multistrategy PSO: t_m iterations are executed to obtain the best subset.

1) In the process of particle motion guided by q strategies, the computational complexity for the selection from the PUS pool is $O(ps \times t_m \times q)$ and for the particle motion is $O(ps \times t_m \times N_A)$.
2) After obtaining the new particles, the solution evaluations need $O(t_m \times O_1)$ and the reference update needs $O(3 \times O_2)$. For extraction and feedback of evolutionary experience to adjust the selection probability pus_j, the computation complexity is approximately $O(t_m \times ps \times q)$.

VI. Conclusion and Future Study

In the article, a novel adaptive multistrategy PSO has been designed for hyperspectral remote sensing image band selection, in view of the local optima stagnation caused by the use of the fixed strategy. In the proposed AMSPSO_BS framework, the quotient of the LD value and MMI serves as the fitness function to evaluate the redundancy between bands, and the subset solution is optimized using the proposed method. The classifiers are finally adopted to classify the multiband subimage made of the selected bands. Experiments were conducted to fully confirm the effectiveness and versatility of AMSPSO_BS. The sensitivity of the parameters and the components of the subsets were also analyzed. The results showed that AMSPSO_BS can obtain a band subset with a low amount of redundant information.

In the future, the hyperspectral remote sensing image band selection problem will be modeled more precisely to obtain a solution with high information, low redundancy, and strong category separability. On the other hand, developing a metric suitable for the application will also be important.

References

[1] S. Thomas et al., "Benefits of hyperspectral imaging for plant disease detection and plant protection: A technical perspective," *J. Plant Diseases Protection*, vol. 125, no. 1, pp. 5–20, Feb. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:09 UTC from IEEE Xplore. Restrictions apply.

[2] B. Wang, X. Zhang, Y. Dong, J. Zhang, J. Zhang, and X. Zhou, "Retrieval of leaf chlorophyll content of paddy rice with extracted foliar hyperspectral imagery," in *Proc. 7th Int. Conf. Agro-Geoinform. (Agro-Geoinform.)*, Aug. 2018, pp. 1–5.

[3] L. Wei et al., "Crops fine classification in airborne hyperspectral imagery based on multi-feature fusion and deep learning," *Remote Sens.*, vol. 13, no. 15, p. 2917, Jul. 2021.

[4] G. Lassalle, "Monitoring natural and anthropogenic plant stressors by hyperspectral remote sensing: Recommendations and guidelines based on a meta-review," *Sci. Total Environ.*, vol. 788, Sep. 2021, Art. no. 147758.

[5] S. Veraverbeke et al., "Hyperspectral remote sensing of fire: State-of-the-art and future perspectives," *Remote Sens. Environ.*, vol. 216, pp. 105–121, Oct. 2018.

[6] S. S. Sawant and M. Prabukumar, "A survey of band selection techniques for hyperspectral image classification," *J. Spectral Imag.*, vol. 9, no. 1, pp. 2040–4565, Jun. 2020.

[7] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[8] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[9] A.-D. Li, B. Xue, and M. Zhang, "Improved binary particle swarm optimization for feature selection with new initialization and search space reduction strategies," *Appl. Soft Comput.*, vol. 106, Jul. 2021, Art. no. 107302.

[10] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.

[11] K. He et al., "A dual global–local attention network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527613.

[12] S. Gu, R. Cheng, and Y. Jin, "Feature selection for high-dimensional classification using a competitive swarm optimizer," *Soft Comput.*, vol. 22, no. 3, pp. 811–822, Feb. 2018.

[13] M. P. Uddin, M. A. Mamun, M. I. Afjal, and M. A. Hossain, "Information-theoretic feature selection with segmentation-based folded principal component analysis (PCA) for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 42, no. 1, pp. 286–321, Jan. 2021.

[14] W. Wong and C. I. Ming, "A review on metaheuristic algorithms: Recent trends, benchmarking and applications," in *Proc. 7th Int. Conf. Smart Comput. Commun. (ICSCC)*, Jun. 2019, pp. 1–5.

[15] P. A. Vikhar, "Evolutionary algorithms: A critical review and its future prospects," in *Proc. Int. Conf. Global Trends Signal Process., Inf. Comput. Commun. (ICGTSPICC)*, Dec. 2016, pp. 261–265.

[16] L. Brezočnik, I. Fister, and V. Podgorelec, "Swarm intelligence algorithms for feature selection: A review," *Appl. Sci.*, vol. 8, no. 9, p. 1521, Sep. 2018.

[17] B. Xue, M. Zhang, W. N. Browne, and X. Yao, "A survey on evolutionary computation approaches to feature selection," *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606–626, Aug. 2016.

[18] B. H. Nguyen, B. Xue, and M. Zhang, "A survey on swarm intelligence approaches to feature selection in data mining," *Swarm Evol. Comput.*, vol. 54, May 2020, Art. no. 100663.

[19] L. Sun and W. Gao, "Method of selecting the best classification bands from hyperspectral images based on genetic algorithm and rough set," *Proc. SPIE*, vol. 3502, pp. 179–184, Aug. 1998.

[20] K. Nagasubramanian, S. Jones, S. Sarkar, A. K. Singh, A. Singh, and B. Ganapathysubramanian, "Hyperspectral band selection using genetic algorithm and support vector machines for early identification of charcoal rot disease in soybean stems," *Plant Methods*, vol. 14, no. 1, p. 86, Oct. 2018.

[21] X. Li and G. Wang, "Optimal band selection for hyperspectral data with improved differential evolution," *J. Ambient Intell. Humanized Comput.*, vol. 6, no. 5, pp. 675–688, Oct. 2015.

[22] W. Ma, X. Zhou, H. Zhu, L. Li, and L. Jiao, "A two-stage hybrid ant colony optimization for high-dimensional feature selection," *Pattern Recognit.*, vol. 116, Aug. 2021, Art. no. 107933.

[23] L. Zhang, Y. Zhong, B. Huang, J. Gong, and P. Li, "Dimensionality reduction based on clonal selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4172–4186, Dec. 2007.

[24] H. Wang, M. Liang, C. Sun, G. Zhang, and L. Xie, "Multiple-strategy learning particle swarm optimization for large-scale optimization problems," *Complex Intell. Syst.*, vol. 7, no. 1, pp. 1–16, Feb. 2021.

[25] Y. Cao, H. Zhang, W. Li, M. Zhou, Y. Zhang, and W. A. Chaovalitwongse, "Comprehensive learning particle swarm optimization algorithm with local search for multimodal functions," *IEEE Trans. Evol. Comput.*, vol. 23, no. 4, pp. 718–731, Aug. 2019.

[26] Z.-H. Zhan, Z.-J. Wang, H. Jin, and J. Zhang, "Adaptive distributed differential evolution," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 4633–4647, Nov. 2020.

[27] D. Gómez and A. Rojas, "An empirical overview of the no free lunch theorem and its effect on real-world machine learning classification," *Neural Comput.*, vol. 28, no. 1, pp. 216–228, Jan. 2016.

[28] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. Int. Conf. Neural Netw. (ICNN)*, vol. 4, Nov. 1995, pp. 1942–1948.

[29] J. C. Bansal, "Particle swarm optimization," in *Evolutionary and Swarm Intelligence Algorithms*, J. C. Bansal, P. K. Singh, and N. R. Pal, Eds. Cham, Switzerland: Springer, 2019, pp. 11–23.

[30] X.-L. Deng, B. Wei, H. Zeng, L. Gui, and X.-W. Xia, "A multi-population based self-adaptive migration PSO," *Acta Electonica Sinica*, vol. 46, no. 8, p. 1858, Aug. 2018.

[31] J.-L. Zhou, J.-S. Wang, Y.-X. Zhang, Q.-S. Guo, H. Li, and Y.-X. Lu, "Particle swarm optimization algorithm with variety inertia weights to solve unequal area facility layout problem," in *Proc. Chin. Control Decis. Conf. (CCDC)*, Aug. 2020, pp. 4240–4245.

[32] Y. Wang, Q. Zhu, H. Ma, and H. Yu, "A hybrid gray wolf optimizer for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527713.

[33] J. Tao, C. Song, D. Song, and X. Wang, "Pan-sharpening framework based on multiscale entropy level matching and its application," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411221.

[34] B. Wei et al., "Multiple adaptive strategies based particle swarm optimization algorithm," *Swarm Evol. Comput.*, vol. 57, Sep. 2020, Art. no. 100731.

[35] Y. Xue, B. Xue, and M. Zhang, "Self-adaptive particle swarm optimization for large-scale feature selection in classification," *ACM Trans. Knowl. Discovery From Data*, vol. 13, no. 5, pp. 1–27, Oct. 2019.

[36] S. Zdiri, J. Chrouta, and A. Zaafouri, "Cooperative multi-swarm particle swarm optimization based on adaptive and time-varying inertia weights," in *Proc. IEEE 2nd Int. Conf. Signal, Control Commun. (SCC)*, Dec. 2021, pp. 200–207.

[37] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[38] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[39] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.

[40] S. Li and H. Qi, "Sparse representation based band selection for hyperspectral images," in *Proc. 18th IEEE Int. Conf. Image Process.*, Sep. 2011, pp. 2693–2696.

[41] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[42] Z. M. Du and J. Feng, "Support vector machine feature selection algorithm based on modified genetic algorithm," *Comput. Eng. Appl.*, vol. 45, no. 29, pp. 28–30, 2009.

[43] Anamika, R. Gupta, and G. Singh, "Binary differential evolution-based feature selection for hand gesture classification," in *Advances in Smart Communication and Imaging Systems*. Singapore: Springer, 2021, pp. 221–231.

[44] B. Xue, M. Zhang, and W. N. Browne, "Particle swarm optimisation for feature selection in classification: Novel initialisation and updating mechanisms," *Appl. Soft Comput.*, vol. 18, pp. 261–276, May 2014.

[45] A. I. Hafez, H. M. Zawbaa, E. Emary, and A. E. Hassanien, "Sine cosine optimization algorithm for feature selection," in *Proc. Int. Symp. Innov. Intell. Syst. Appl. (INISTA)*, Sinaia, Romania, Aug. 2016, pp. 1–5.

[46] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe. (Sep. 2015). *220 Band Aviris Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3*. [Online]. Available: https://purr.purdue.edu/publications/1947/

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[47] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[48] C. Zhong, L. Li, and F. Bu, "Study of modified band selection methods of hyperspectral image based on optimum index factor," *Proc. SPIE*, vol. 9299, pp. 246–251, Nov. 2014.



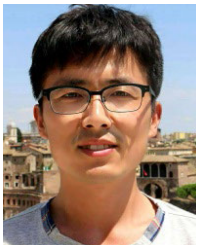
Yuting Wan (Member, IEEE) received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2017, the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

He is currently a Post-Doctoral Researcher with Wuhan University. His major research interests include remote sensing image processing, evolutionary computation, multiobjective optimization, and neural architecture search.



Chao Chen (Student Member, IEEE) received the B.S. degree from Electronic Information School, Wuhan University, Wuhan, China, in 2022, where he is currently pursuing the M.S. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing.

His major research interests include hyperspectral remote sensing image compression, evolutionary computation, and multiobjective optimization.



Ailong Ma (Member, IEEE) received the B.S. degree from the China University of Petroleum, Qingdao, China, in 2010, the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

He is now working as an Research Associate with Wuhan University. His major research interests include remote sensing image processing, evolutionary computing, and deep learning.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

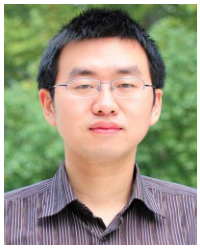
He was a Principal Scientist for the China State Key Basic Research Project from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He is a "Chang-Jiang Scholar" Chair Professor with the Ministry of Education of China, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He has authored more than 700 research papers and five books. He is the Institute for Scientific Information (ISI) highly cited author. He is the holder of 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He is currently serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Xunqiang Gong received the Ph.D. degree in photogrammetry and remote sensing from Southwest Jiaotong University, Chengdu, China, in 2017.

From 2015 to 2017, he was a Joint Training Ph.D. Student with the Technical University of Berlin, Berlin, Germany. He is an Associate Professor with the Key Laboratory of Mine Environmental Monitoring and Improving around Poyang Lake, Ministry of Natural Resources, East China University of Technology, Nanchang, China. At the same time, he is a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, and Jiangxi Academy of Eco-Environmental Sciences Research and Planning, Nanchang. He has authored more than 40 research papers in different journals and conferences, and obtained 20 research projects supported by various departments. His research interests include intelligent processing and application of remote sensing images.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis, and Applications of Remote Sensing (RSIDEA) Research Group. He has authored more than 150 research papers in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *International Journal of Remote Sensing*.

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